From a single LLM call to a decision system

Our contribution is the complete workflow around extraction — not just a different prompt.

STANDARD APPROACH

ONE PASS

01 READ
Text from PDF

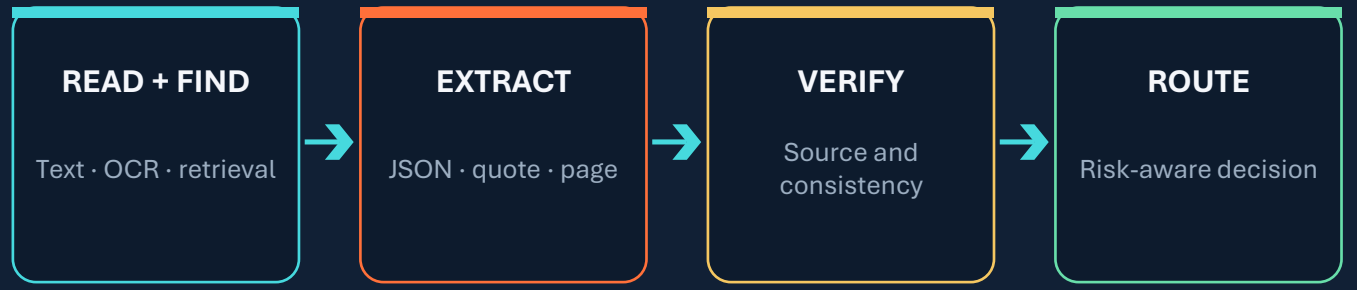
02 EXTRACT
One model pass

03 DECIDE
Trust score → accept

Main limitation: the model score is not proof.

OUR DOCUMENT INTELLIGENCE PIPELINE

EVIDENCE FIRST



ACCEPT
Verified + above threshold

RECHECK
Uncertain → reread

HUMAN REVIEW
Missing proof / high risk

OUR CORE IDEA A model proposes values. Evidence + calibration determine whether we trust them.

WHAT WE BUILT

How the system reads, finds and extracts

Different document types use different reading paths, but all produce structured candidate fields.

CONTRACTS

Long text + 41 clause types

Paragraph retrieval; weighted clause cues and learned terms

SCANNED FORMS

OCR / image-based reading

Vision path for PDFs without a usable text layer

INVOICES

Dates, totals, identifiers

Schema-defined fields, values and source evidence

01 DOCUMENT READER

Text layer · OCR · page images

→

02 RETRIEVAL

Targeted chunks per field / clause

→

03 LLM EXTRACTION

Schema-guided JSON + evidence

EACH EXTRACTED FIELD CARRIES

VALUE	EVIDENCE	PAGE
Candidate answer	Verbatim quote	Source location

DATA DRIVEN DESIGN CHOICES

28.4% contracts exceed 60k characters

41.3% ExtractBench PDFs lack usable text

Why this matters: retrieve evidence instead of truncating long text or ignoring scans.

The layer that makes extraction safer

Every candidate passes evidence checks, optional model verification, and a calibrated risk policy.

01 PROVE

Quote, span and page checks
Hard failure → review

02 VERIFY

Check value against source
Model slot is configurable

03 CALIBRATE

Fit using development data
Freeze threshold and signature

ADAPTIVE FIELD-LEVEL ROUTING — Only uncertain fields trigger extra model calls

ACCEPT

Strong evidence + calibrated score

Output accepted field and value.

RECHECK

Gray-zone / ambiguous answer

Independent reread + challenge, then rescore.

HUMAN REVIEW

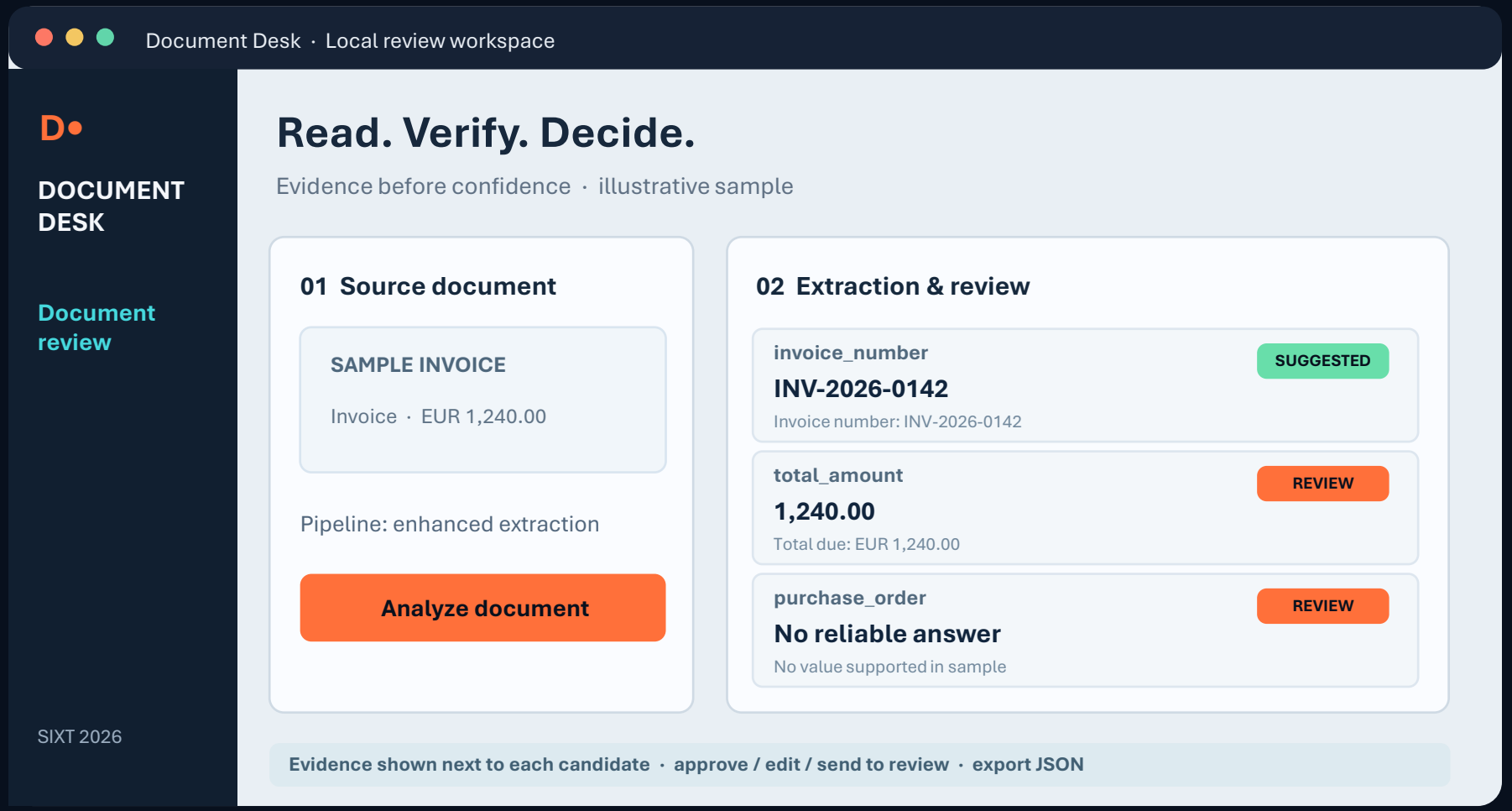
Missing proof / unsafe field

Inspect, edit or approve in Document Desk.

RISK POLICY — **1 human review vs 100 for a wrong auto-fill** — break-even >99.01% precision

Document Desk: where humans remain in control

A working local interface connects document upload, extraction evidence, review actions and JSON export.



DEMO HIGHLIGHTS

1 **Upload**

PDF, image or TXT + extraction schema

2 **Inspect**

Value, evidence, page, status

3 **Take action**

Approve, edit, review, export JSON

ONE REAL SCREENSHOT

Illustrative UI schematic from project sample data — not a live model result.

What we can prove today — and what comes next

Separate verified engineering checks and offline safety behavior from unmeasured live-model quality.

716

PREPARED DOCUMENTS

540 dev + 176 public test

29,077

LABELED FIELDS

22,115 dev + 6,962 test

47 / 47

SOFTWARE CHECKS PASSED

21 pipeline + 7 app + 19 scorer

0

UNSAFE AUTO-ACCEPTS

Offline all-review policy

MEASURED OFFLINE PUBLIC CHECK

176 documents · 6,962 fields

0 auto-accepted · 0% net saving

All fields routed to review because no matching live calibration was available.
This is a safety result, not an accuracy gain.

NEXT: FAIR LIVE COMPARISON

1. Live model extraction on development documents
2. Calibrate and freeze each system's threshold
3. **Evaluate baseline vs enhanced on untouched test**

Report: precision · accepted fields · wrong accepts · coverage · net saving · runtime

THE TAKEAWAY **Automate only when the evidence — and the economics — justify it.**